

SatsPath Implementations

This document maps the SatsPath Protocol v1 specification to the current repository implementation.

The implementation must be understood as a protocol stack, not as a single P2P system. P2P is one transport implementation among several.

Repository Layout

<code>crates/satspath-core</code>	protocol data types, signatures, resolvers, validation
<code>crates/satspath-router</code>	quote response contract and route selection
<code>crates/satspath-cli</code>	command-line reference client
<code>crates/satspathd</code>	local daemon and HTTP API
<code>sdk/satspath-p2p</code>	optional Pear/Holepunch transport
<code>docs/</code>	protocol and operational documentation

Core Protocol Types

Implemented in:

`crates/satspath-core/src/profile.rs`

Spec mapping:

Spec object	Rust type
PaymentProfile	PaymentProfile
SignedPaymentProfile	SignedPaymentProfile
PaymentMethod.Lightning	PaymentMethod::Lightning
PaymentMethod.Onchain	PaymentMethod::Onchain
PaymentMethod.Ark	PaymentMethod::Ark
Invite	Invite, InviteRecord

Signature and Safety Validation

Implemented in:

`crates/satspath-core/src/crypto.rs`
`crates/satspath-core/src/validation.rs`

Responsibilities:

- Generate protocol identity keypairs.
- Sign public profiles.
- Verify signed profiles.
- Compute identity fingerprints.
- Reject malformed public keys.
- Reject private material in public protocol objects.
- Validate Lightning addresses, Bitcoin addresses, Ark URLs, and compressed pubkeys.

Protocol identity keys are not wallet spending keys.

Resolver Implementations

Implemented in:

`crates/satspath-core/src/resolver.rs`
`crates/satspath-core/src/resolvers/`

crates/satspath-core/src/registry.rs
crates/satspath-core/src/peer_registry.rs

Current resolver surfaces:

Resolver	File	Status
Local registry	registry.rs	Active
Local peer registry	peer_registry.rs	Active local storage
BIP-353	resolvers/bip353.rs, bip353.rs	Resolver and DNS primitives; strict DNSSEC fails closed without validator
HTTPS	resolvers/http.rs	Active
Nostr	resolvers/nostr.rs	Active NIP-05 + kind 30078 resolver
Platform	resolvers/platform.rs	Scaffold

Resolver chain behavior is implemented by `ChainResolver`.

Quote Response and Routing

Implemented in:

crates/satspath-router/src/quote_response.rs
crates/satspath-router/src/router.rs
crates/satspath-router/src/fees.rs
crates/satspath-router/src/lightning.rs

Spec mapping:

Spec behavior	Implementation
Resolve identifier	quote_inner + ProfileResolver
Verify profile	verify_signed_profile
Check expiry	check_profile_expiry
Select route	select_route, select_route_with_fees
Build payment payload	build_qr_payload
Stable response	QuoteResponse

QuoteResponse status values:

ok
not_registered
no_route
invalid_signature

These values are the UI and API contract.

Daemon Implementation

Implemented in:

crates/satspathd/src/main.rs

The daemon exposes the protocol over a local HTTP API:

Endpoint	Purpose
GET /health	Liveness
GET /v1/status	Local daemon/protocol status
GET /v1/node	Aggregate status, profile, peers, connections
GET /v1/profile	Local wallet profile state
PUT/POST /v1/profile	Create or update local public profile
POST /v1/profile/methods	Update receive methods
POST /v1/resolve	Resolve a local profile
POST /v1/quote	Protocol quote response
POST /v1/pay	Wallet handoff using protocol quote response
POST /v1/dns/resolve	BIP-353/DNS resolution

/v1/pay does not move funds. It returns a wallet handoff containing a public payment payload and QR SVG.

CLI Implementation

Implemented in:

`crates/satspath-cli/src/`

Important commands:

Command area	Protocol role
<code>register</code>	Create signed public profile
<code>wallet</code>	Manage local receive profile
<code>quote</code>	Produce quote response
<code>pay</code>	Preview/handoff flow
<code>dns</code>	BIP-353 resolver tooling
<code>peer export/import</code>	Manual transport for signed profiles

The CLI is a reference client for local development and protocol testing.

P2P Transport Implementation

Implemented in:

`sdk/satspath-p2p/`

This SDK is an optional transport. It publishes and resolves signed profiles over Pear/Holepunch. It must be treated as a resolver transport, not the whole protocol.

Conformance requirements:

- Return `SignedPaymentProfile` objects.
- Verify before import or route use.
- Never treat peer connectivity as profile ownership.
- Never carry wallet private material.

Wire behavior is documented in `wire_p2p.md`.

Current Gaps

Known v1 implementation gaps:

- DNSSEC strict mode needs a local DNSSEC-validating resolver to fully trust BIP-353 on mainnet.
- Nostr publishing is not yet exposed as a Rust CLI command; use a Nostr client to publish the kind 30078 event. Platform resolvers are scaffolds.
- Resolver provenance is not yet carried through `ProfileResolver`, so signed-profile quote responses mark `identifier_verified: false` unless a direct BIP-353 preview provides DNSSEC validation.
- Mainnet payment execution is intentionally not implemented.
- P2P wire envelopes should be formalized in the SDK to match `wire_p2p.md`.
- Method ownership proofs exist in core but need broader resolver integration.

Conformance Checklist

An implementation in this repo should be considered conformant when it:

- Uses `SignedPaymentProfile` for receiver data.
- Verifies signatures before routing.
- Applies expiry checks.
- Rejects unsafe private material.
- Produces the four-status `QuoteResponse`.
- Keeps resolver transports separate from protocol verification.
- Treats Pear/Holepunch as optional transport.
- Returns wallet handoff data instead of executing mainnet payments.